

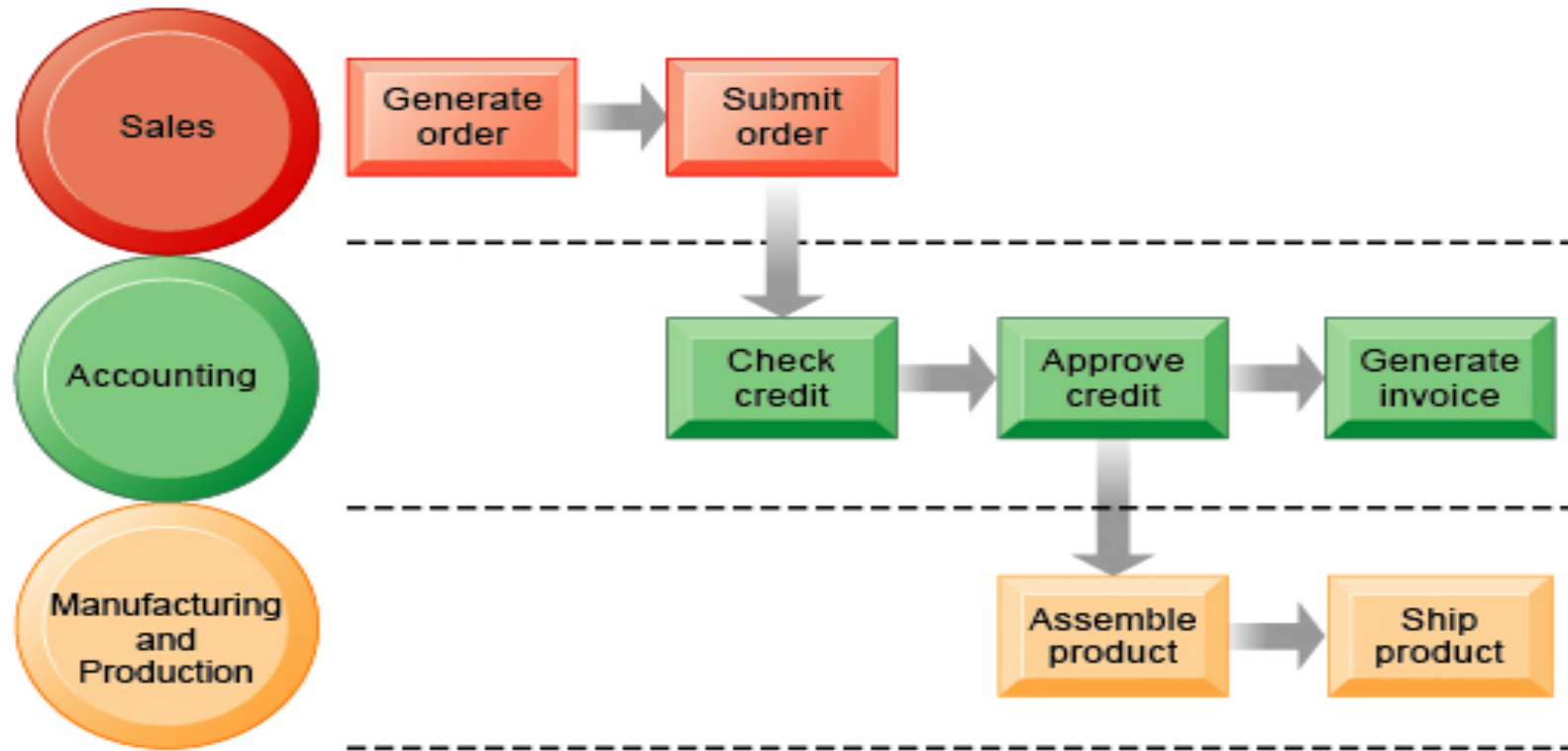
# Chapter 5 : Enterprise Information Systems



# Cross-Functional Business Processes

- Cross-functional business process means when different departments within the same company work hand in hand to achieve a common goal.
- For example, the sales department may work with the shipping department. The marketing department may work with the sales department
- Transcend boundary between sales, marketing, manufacturing, and research and development
- Group employees from different functional specialties to a complete piece of work
- Example – Order Fulfilment Process

# Cross-Functional Business Processes

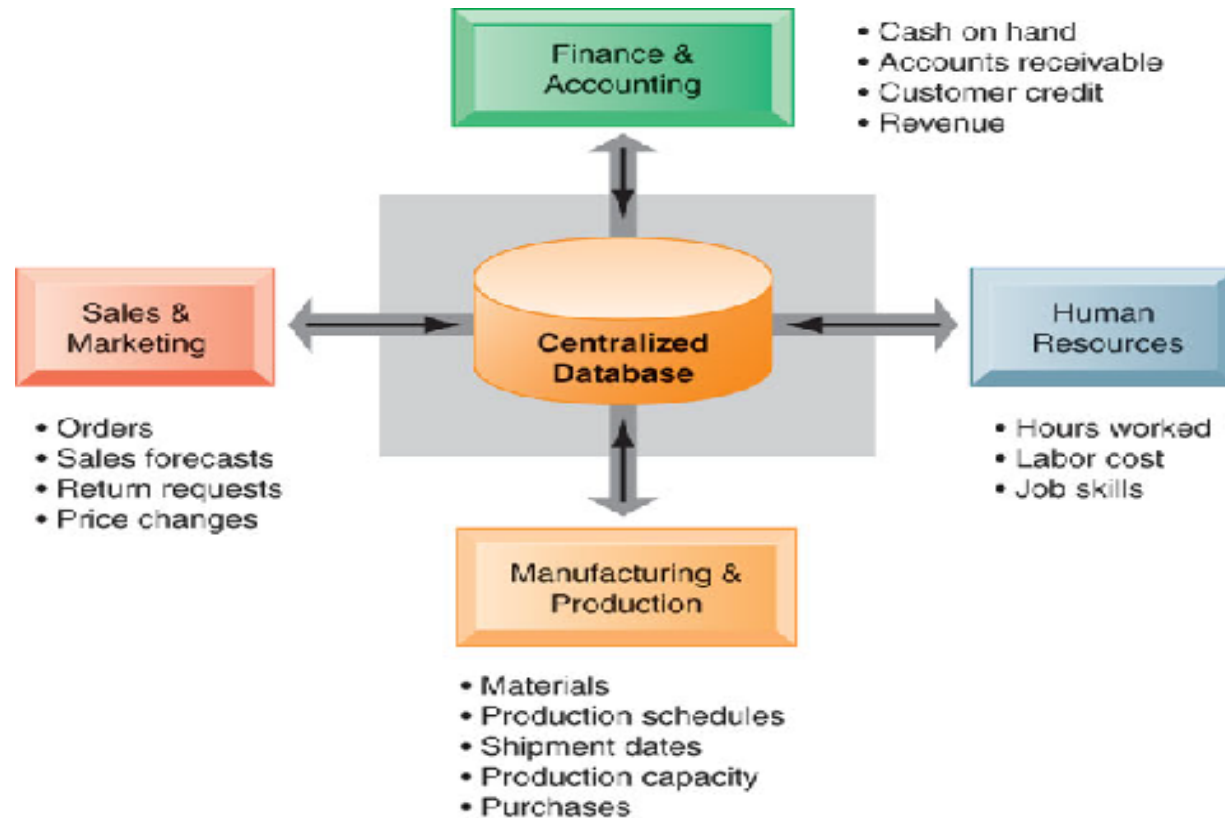


The order fulfillment process

# Enterprise Systems and Applications

- Enterprise systems feature a set of integrated software modules and a central database that enables data to be shared by many different business processes and functional areas throughout the enterprise.
- Enterprise systems are large-scale software packages that are able to track and control all of the complex operations of a business.
- These systems are used as a central command hub to help automate the business and make reporting and decision making easier.
- Companies implementing this software must first select the functions of the system they wish to use and then map their business processes to the predefined business processes in the software

# Enterprise Systems Architecture



# Enterprise Resource Planning (ERP)

- ERP a process used by companies to manage and integrate the important parts of their businesses.
- Many ERP software applications exist to help companies implement resource planning by integrating all of the processes it needs to run a company with a single system
- Provides single information system for organization-wide coordination and integration of key business processes
- Models and automates many business processes

# ERP Model



# Business Processes Supported By Enterprise Systems

- **Financial and accounting processes** - including general ledger, accounts payable, accounts receivable, fixed assets, cash management and forecasting, product-cost accounting, cost-center accounting, asset accounting, tax accounting, credit management, and financial reporting.
- **Human resources processes** - including personnel administration, time accounting, payroll, personnel planning and development, benefits accounting, applicant tracking, time management, compensation, workforce planning, performance management, and travel expense reporting.
- **Manufacturing and production processes** - including procurement, inventory management, purchasing, shipping, production planning, production scheduling, material requirements planning, quality control, distribution, transportation execution, and plant and equipment maintenance.
- **Sales and marketing processes** - including order processing, quotations, contracts, product configuration, pricing, billing, credit checking, incentive and commission management, and sales planning.

# Business Value of ERP

- Enterprise systems provide value both by increasing operational efficiency and by providing firm-wide information to help managers make better decisions.
- Large companies with many operating units in different locations have used enterprise systems to enforce standard practices and data so that everyone does business the same way worldwide
  - Coca Cola, for instance, implemented a SAP enterprise system to standardize and coordinate important business processes in 200 countries.
- Enterprise systems help firms respond rapidly to customer requests for information or products by integrating order, manufacturing, and delivery data, manufacturing is better informed about producing only what customers have ordered, procuring exactly the right amount of components or raw materials to fill actual orders, staging production, and minimizing the time that components or finished products are in inventory.

# Business Value of ERP

- Enterprise systems provide much valuable information for improving management decision making. Corporate headquarters has access to up-to-the minute data on sales, inventory, and production and uses this information to create more accurate sales and production forecasts.
- Enterprise software includes analytical tools for using data captured by the system to evaluate overall organizational performance.
- Enterprise system data have common standardized definitions and formats that are accepted by the entire organization. Performance figures mean the same thing across the company.
- Enterprise systems allow senior management to easily find out at any moment how a particular organizational unit is performing, determine which products are most or least profitable, and calculate costs for the company as a whole.

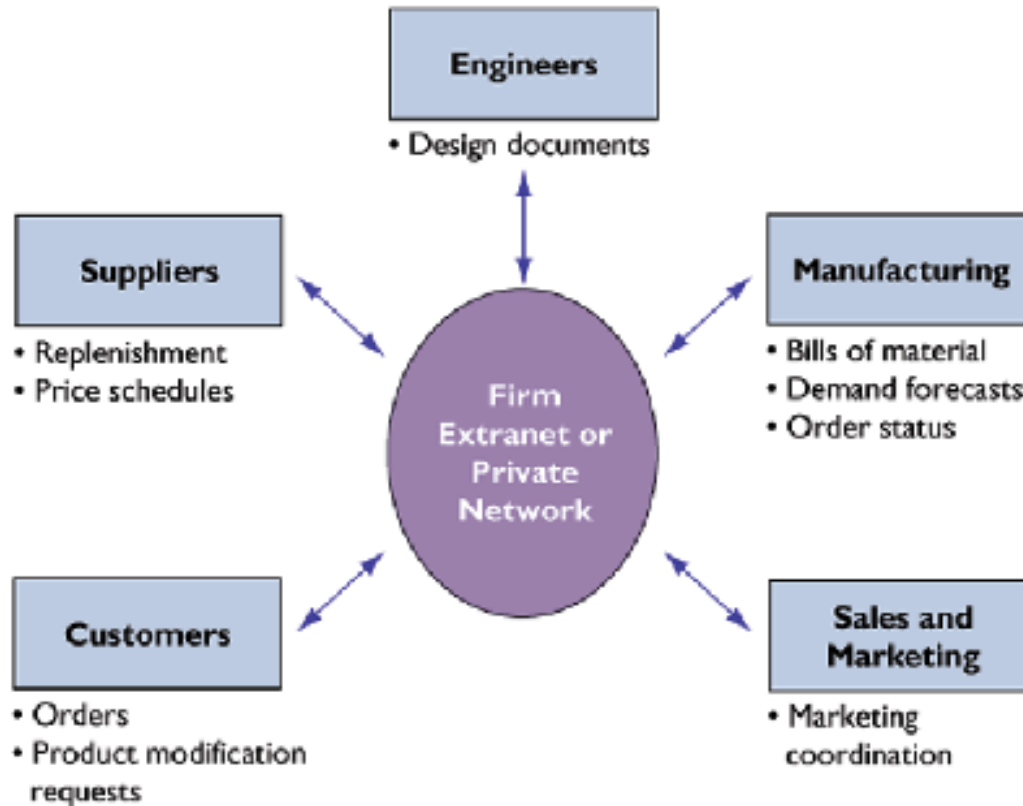
# ERP and Collaborative Commerce

- ERP was confined to only big businesses when it started, however, over the years, small and medium size companies too adopted ERP implementation.
- Corporations or organizations that manage a lot of data flowing can adopt ERP systems for better data management. There are different ERP solutions designed for different processes that run through in organizations.
- Some of the major collaborative commerce platforms enabled and supported by ERP include
  - ❑ Supply Chain Management
  - ❑ Customer Relations Management
  - ❑ Knowledge Management

# Collaborative Commerce

- Uses digital technologies to enable multiple organizations to collaboratively design, develop, build, move, and manage products
- Increases efficiencies in reducing product design life cycles, minimizing excess inventory, forecasting demand, and keeping partners and customers informed
- Collaboration between partners to formulate demand forecasts, develop production plans, coordinate shipping, warehousing, stocking
- Utilizes web-enabled networks for coordinating trans organizational business processes

# Collaborative Commerce



# Supply Chain Management Systems

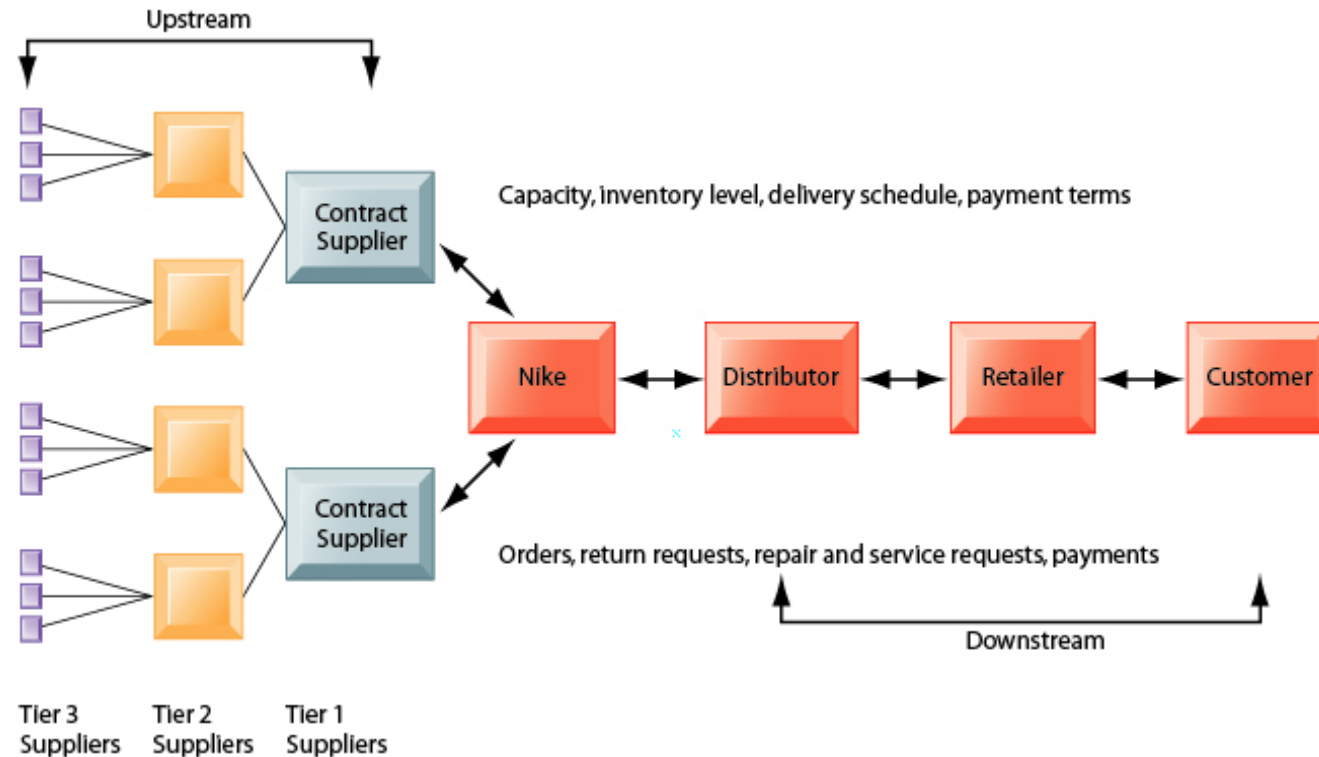
- A supply chain management (SCM) system is a set of software solutions that manages and oversees the flow of goods, data, and finances as a product or service moves from point of origin to its final destination. Supply chain activities encompass procurement, product lifecycle management (PLM), supply chain planning (SCP), logistics and order management.
- A complete, end-to-end supply chain management system includes the material handling and software packages for all the parties who work together to create the product, fulfill orders, and keep track of information including suppliers, manufacturers, wholesalers, transportation providers, logistics providers, and retailers.

# Supply Chain Management Systems

- **Supply Chain Management Systems (SCM):** Automate flow of information between firm and suppliers to optimize production and delivery
- **Supply Chain Management:** Close linkage of activities involved in buying, making, moving a product
- **Supply Chain:** Network of organizations and business processes for production and distribution of products

# Supply Chain Management Systems

Nike's Supply Chain Model



# Supply Chain Management Systems

- How information systems facilitate supply chain management
  - ❑ Decide when and what to produce, store, and move
  - ❑ Rapidly communicate orders
  - ❑ Track the status of orders
  - ❑ Check inventory availability and monitor inventory levels
  - ❑ Reduce inventory, transportation, and warehousing costs
  - ❑ Track shipments
  - ❑ Plan production based on actual customer demand
  - ❑ Rapidly communicate changes in product design

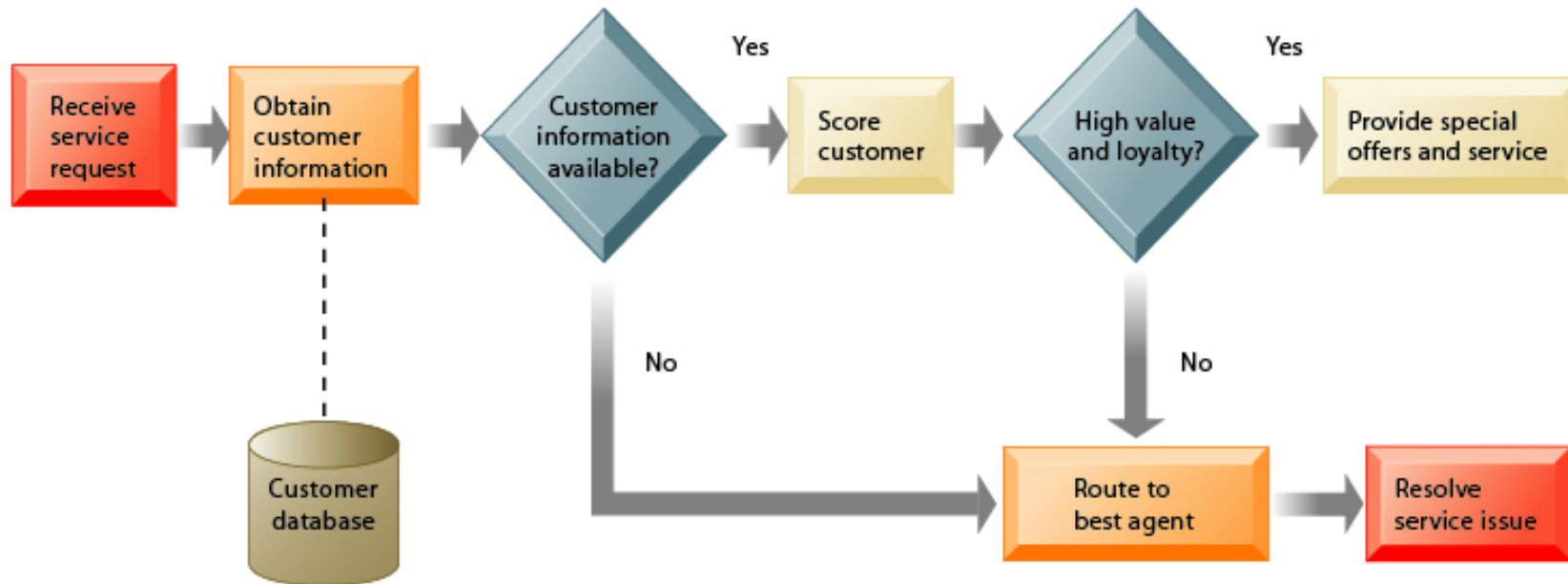
# Customer Relationship Management (CRM) Systems

- Customer relationship management (CRM) systems capture and integrate customer data from all over the organization, consolidate the data, analyze the data, and then distribute the results to various systems and customer touch points across the enterprise
- A Customer Relationship Management (CRM) system enhances a business's ability to cultivate and nurture customer relationships. The system provides a unified approach to organizing customer information while also streamlining processes.
- Traditionally, due to the customer focus, CRM has been used primarily as a sales and marketing tool. However, in recent years, certain CRMs have evolved to play a more central role, expanding beyond sales and marketing into almost every department.

# CRM Systems Features and Capabilities

- **Sales Force Automation** : provides users an easy to follow process to navigate the sales cycle. Tracking customer interaction, analyzing sales forecasts, visualizing sales trends and automating business tasks are a few examples that help to maximize productivity.
- **Marketing Automation**: help to reduce repetitive tasks. With Marketing Automation, a potential customer might get an automated email or tweet after requesting information from the company website
- **Mobile with Offline Access**: allows users the flexibility to utilize CRM regardless of their internet connectivity.
- **Field Service Automation** : ability to locate a technician in real time, users can help route the next closest call, improving response time as well as increasing the number of daily calls serviced

# Customer Loyalty Management Process Map



This process map shows how a best practice for promoting customer loyalty through customer service would be modeled by customer relationship management software. The CRM software helps firms identify high-value customers for preferential treatment.